



SAMRAT POLY RESINS

Manufacturer of Coating & Composite Resins · Doraha, Ludhiana, Punjab, India

TECHNICAL DATA SHEET

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UV STABILIZED SHEET GRADE YELLOW RESIN

Pre-Pigmented Orthophthalic Unsaturated Polyester Resin · UV-Stabilised · Sheet and Panel Production

PRODUCT DESCRIPTION

UV Stabilized Sheet Grade Yellow Resin is a pre-pigmented unsaturated polyester resin manufactured by Samrat Poly Resins for FRP sheet and panel production where the finished part is exposed to sunlight. It is supplied as a yellow liquid with the pigment already dispersed at the factory, so colour does not have to be added on the shop floor, and it carries a UV-stabilised formulation intended to improve resistance to prolonged outdoor exposure and weathering. It is a distinct grade from the standard Sheet Grade Yellow Resin, which is supplied for general sheet and panel manufacturing without a UV-stabiliser package.

KEY FEATURES

- Pre-pigmented yellow resin — colour dispersed at the factory, not added at the laminating stage
- UV-stabilised formulation intended for prolonged outdoor exposure
- Formulated for FRP sheet moulding and lamination processes
- Orthophthalic unsaturated polyester base
- Room-temperature cure with MEKP catalyst and cobalt octoate accelerator

RECOMMENDED APPLICATIONS

- FRP sheet and panel production for outdoor service
- Coloured roofing and cladding sheet
- Sheet moulding and lamination
- Exterior-exposed coloured FRP components

PRODUCT SPECIFICATION

Working draft — the test conditions marked below are pending confirmation.

IDENTITY AND SUPPLY		
Resin Type	Orthophthalic Unsaturated Polyester Resin	supplied by Samrat Poly Resins
Grade	UV Stabilized Sheet Grade Yellow Resin	supplied by Samrat Poly Resins
Form	Pre-pigmented yellow liquid resin	supplied by Samrat Poly Resins
Supplied Colour	Yellow — factory master yellow shade	supplied by Samrat Poly Resins
Acceleration Status	Non-accelerated	accelerator is added at the point of use
Packaging	225 kg industrial drum	supplied by Samrat Poly Resins
Country of Origin	India	supplied by Samrat Poly Resins
LIQUID RESIN PROPERTIES		
Viscosity	280 cPs	at 25 °C; spindle and speed — to be confirmed
Gel Time	6.5 minutes	at 25 °C; catalyst and accelerator levels used in this determination — to be confirmed
Specific Gravity	1.10 g/cm ³	at 25 °C
Styrene Content	35% by weight	determination method — to be confirmed
CURE SYSTEM		
Curing System	MEKP catalyst with cobalt octoate accelerator	supplied by Samrat Poly Resins
Recommended Catalyst Dosage	MEKP 1.5% by weight of resin	supplied by Samrat Poly Resins
Recommended Accelerator Dosage	Cobalt octoate 0.4% by weight of resin	cobalt metal content this dosage assumes — to be confirmed
UV STABILISATION		
Stabiliser System	UV absorber with HALS light stabiliser	supplied by Samrat Poly Resins
UV Absorber Type	Benzotriazole type	supplied by Samrat Poly Resins
HALS	Hindered amine light stabiliser	supplied by Samrat Poly Resins
Cure Mechanism	UV-stabilised, not UV-light-curing	the stabiliser protects the cured laminate; it takes no part in the cure
WEATHERING, AS SUPPLIED		
Test Method	ISO 4892-2 xenon arc	exposure cycle, irradiance, filter and black-standard temperature — to be confirmed
Alternative Method	ASTM G155 xenon arc	named as a comparable alternative method; not a second exposure that was performed

Exposure Duration	1,000 hours	supplied by Samrat Poly Resins
Colour Change	ΔE 3–5	colour measurement standard — to be confirmed
Gloss Retention	80%	gloss geometry and measurement standard — to be confirmed
Tensile Strength Retention	85%	tensile test standard and specimen form — to be confirmed
SAFETY AND STORAGE		
Flash Point	32 °C	closed cup
Shelf Life	2 months	in original tightly closed containers at the storage conditions below
Storage Temperature	15–25 °C, maximum recommended 25 °C	supplied by Samrat Poly Resins
Storage Conditions	Original tightly closed containers in a cool, dry, shaded, well-ventilated area. Protect from direct sunlight, UV exposure, heat, sparks, ignition sources and moisture.	supplied by Samrat Poly Resins

Values in this table are supplied by Samrat Poly Resins for this grade. They are typical values for the product as manufactured, not guaranteed limits. Conditions marked “to be confirmed” are test conditions not yet stated; they are shown as outstanding rather than assumed. This grade is a UV-stabilised polyester resin: it cures with MEKP and a cobalt octoate accelerator, and is not a UV-light-curing resin.

GRADE SELECTION

Where the finished sheet is translucent and optical clarity matters rather than colour, Roof Light Sheet Grade Resin is the grade to review. For general interior sheet and panel manufacturing without outdoor exposure, Sheet Grade Yellow Resin is the standard route. Confirm the intended sheet product, exposure condition and processing route before selecting a grade.

SAFETY & HANDLING

Handle with appropriate personal protective equipment and adequate ventilation. Unsaturated polyester resins contain styrene, which is flammable. Keep away from heat, sparks and open flame. **Never mix the MEKP catalyst directly with the cobalt accelerator. Add each separately into the resin — direct contact between the two can react violently.** Refer to the applicable Safety Data Sheet (SDS) before handling.

Disclaimer: The information above is based on data published by the manufacturer at the time of issue and is believed accurate. Values shown are typical and are not to be interpreted as guaranteed specifications; properties may vary within normal manufacturing tolerances between batches. Users are responsible for determining suitability for their intended application and should confirm current specifications directly with Samrat Poly Resins prior to use.